Quantification: 682_z08
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	8
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2599.88	2671.5	1127.64	px^2
major_axis_length_px	lacuna shape	103.523	92.8874	29.9285	px
minor_axis_length_px	lacuna shape	33.6052	36.6597	6.7292	px
aspect_ratio	lacuna shape	3.087	2.9214	0.6438	unitless
eccentricity	lacuna shape	0.9401	0.9395	0.0221	unitless
solidity	lacuna shape	0.8296	0.873	0.1073	unitless
roots_count	headline	10.5	9	4.504	count
ring_length_r30_px	headline	422.875	368.5	133.175	px
ring_length_r60_px	reported	1024.38	889	346.612	px
owned_length_px	ownership-dependent	2735.34	1863.43	2088.96	px
edge_count	ownership-dependent	125.625	86	102.215	count
mean_edge_length_px	ownership-dependent	22.5629	22.6394	2.7001	px
perimeter_px	normalised	275.426	240.43	89.7432	px
ring_area_r30_px2	normalised	10379.6	9533.5	2334.04	px^2
ring_area_r60_px2	normalised	24427.5	24429.5	5136.87	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9949	1	0.0144	unitless
ring_density_r30	normalised	0.0405215	0.0390801	0.00643539	px^-1
ring_density_r60	normalised	0.0416417	0.0415977	0.00786325	px^-1
roots_per_100px_perimeter	normalised	3.7643	3.936	0.6189	per 100 px
ring_attached_length_r30_px	network v2	351.625	305	139.588	px
ring_attached_length_r60_px	network v2	809.5	668.5	395.519	px
ring_length_w_r30_px	network v2	477.09	412.605	144.038	px
ring_length_w_r60_px	network v2	1157.94	991.589	380.693	px
sholl_crossings_r10	network v2	10.875	9.5	5.0832	count
sholl_crossings_r20	network v2	14	12.5	4.0356	count
sholl_crossings_r30	network v2	16.5	14	5.3452	count
ring_width_mean_r30_px	width	6.2246	6.2912	0.4579	px

Per field
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measure	value	units
analysed_area_px2	1.01791e+06	px^2
total_skeleton_length_px	32768	px
canalicular_length_density_per_px	0.0321915	px^-1
junction_count	672	count
junction_density_per_px2	0.00066018	px^-2
lacuna_count	10	count
lacunae_per_px2	9.82e-06	px^-2
skeleton_component_count	309	count

mean_component_length_px	106.045	px
median_component_length_px	19	px
field_length_density_w_per_px	0.036215	px^-1
field_density_without_flagged_per_px	0.033537	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	6.7319	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	31623	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	9321	210.577	69.7066	3.0209	0.9436	0.684	1.3112	31.8256	428.162	True
2	2563	90.1026	36.7546	2.4515	0.913	0.9144	1.5381	342.687	87.595	False
3	3496	113.895	40.712	2.7976	0.9339	0.8552	-1.2503	434.254	701.917	False
4	4123	159.751	36.5648	4.369	0.9735	0.7865	1.4743	433.966	210.1	False
5	1521	89.609	30.1271	2.9744	0.9418	0.641	-1.2077	520.501	556.951	False
6	862	61.0771	21.2942	2.8683	0.9373	0.7077	-1.4443	545.484	438.164	False
7	547	50.4784	17.0874	2.9541	0.941	0.8116	-0.4973	672.987	1013.45	True
8	3601	127.976	37.0473	3.4544	0.9572	0.8909	1.4857	713.926	152.006	False
9	1853	92.0913	27.1748	3.3889	0.9555	0.9155	-1.2655	723.731	828.852	False
10	2780	93.6835	39.1668	2.3919	0.9084	0.926	-1.1692	908.613	727.092	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	12	391	939	950.671	45	21.126	7.3833
2	9	369	912	1039.42	48	21.6547	6.4993
3	13	418	1040	1410.63	56	25.1898	6.3984
4	20	694	1734	6115.34	317	19.2913	7.0022
5	8	357	753	2181.28	89	24.5088	6.184
6	7	368	866	1973.2	104	18.973	5.5994
7	3	130	379	261.363	7	37.3375	5.269
8	12	547	1341	6024.12	255	23.624	6.4083
9	9	344	793	1753.66	83	21.1284	6.0175
10	6	286	756	1385.05	53	26.1331	5.6877

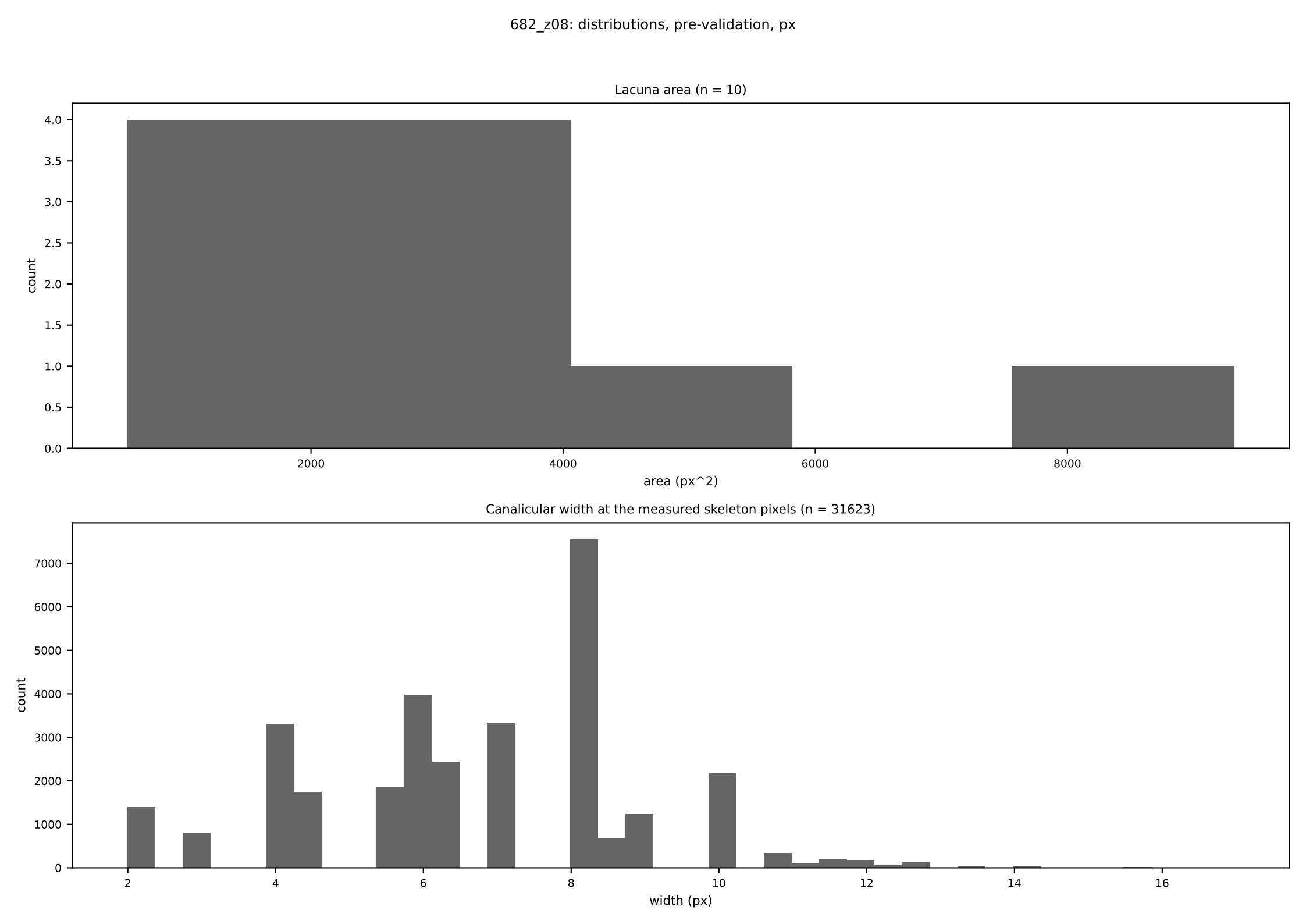
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	769.541	14367	29344	0.6579	0.6229	0.0272152	0.0319997	1.5594
2	234.894	9374	20769	1	0.9593	0.0393642	0.0439116	3.8315
3	313.948	11274	26474	1	1	0.0370765	0.0392838	4.1408

4	456.961	14983	32605	1	1	0.0463192	0.053182	4.3767
5	233.238	9202	19621	1	1	0.0387959	0.0383773	3.43
6	166.125	7387	17424	1	1	0.0498173	0.0497016	4.2137
7	122.604	3917	10590	0.6212	0.5802	0.0331887	0.0357885	2.4469
8	329.522	12020	29668	1	1	0.0455075	0.0452002	3.6416
9	222.752	9104	23858	1	1	0.0377856	0.0332383	4.0404
10	245.965	9693	25001	1	1	0.0295058	0.0302388	2.4394

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	313	666	432.534	1020.64	14	12	11
2	323	727	415.806	1019.77	11	11	15
3	393	736	475.617	1185.92	18	12	13
4	631	1650	763.458	1946.52	18	22	26
5	249	552	409.404	869.193	5	13	13
6	287	610	405.421	963.411	6	12	13
7	80	200	153.924	450.529	3	6	7
8	462	1138	622.458	1490.19	13	18	24
9	259	594	395.818	926.732	8	14	15
10	209	469	328.735	861.808	8	10	13


682_z08: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 31623)
count
width (px)